



COURSE SYLLABUS

Bachelor of Science in Information Technology First Semester, SY 2026-2027

COLLEGE	:	Information Technology & Data Science Department Sarmiento Campus
PROGRAM	:	Bachelor of Science in Information Technology with Specialization in Data and Business Analytics
COURSE CODE	:	Cap 401
COURSE TITLE	:	Capstone Project and Research 2
CREDIT UNITS	:	3 units
PRE-REQUISITE	:	CAP301 – Capstone Project and Research 1
FACULTY	:	Joann G. Perez, Elenita T. Capariño
CONSULTATION HOURS:		3

COURSE DESCRIPTION:

Capstone Project and Research 2 is a semi-independent or directed-study course where students are required to complete a final capstone project based on the proposal developed in Capstone Project and Research 1. The course is designed to allow the students to apply the concepts and skills gained throughout the program by requiring a Capstone project that focuses on information systems development, web and mobile development integrating data and business analytics and implementation as a business solution for an industry need. The students are required to present a functional system together with its full documentation.

Bisyon

Ang Bulacan State University ay isang progresibong institusyong lumilikha ng kaalaman na kinikilala sa daigdig sa pinakamahasag na pagtuturo, nangungunang mga pananaliksik, at maagap na pagtugon sa pangangailangan ng komunidad.

Misyon

Ang Bulacan State University ay may hangaring humubog ng mga propesyonal na mahuhusay. etikal, at handang maglingkod, na nakapag-aambag sa patuloy na pag-unlad ng bayan.



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Core Values: ALAB BulSU!

Agap sa Pagbabago
Lingap sa Kapwa at Sangnilikha
Agapayan sa paggawa
Bayanihan sa banyuhay

The BulSU Ideal Graduates Attributes (BIG A) reflect the graduate’s capacity as:

- A. Makadiyos at Makatao (Ethical and Socially Responsible);
- B. Matalino at Malikhain (Critical and Creative Thinker);
- C. Mapanagutan at Maasahan (Accountable and Dependable Professional); and
- D. Makabansa at Makabuluhan ang Gawa (Service-Oriented and Nationalistic)

Program Educational Objectives (PEO)

Program Educational Objectives (PEO)	University Mission			
	BIG A-a	BIG A-b	BIG A-c	BIG A-d
PEO1. Graduates thrive as industry professionals or technopreneurs by leveraging advanced technical expertise, innovative problem-solving, and strategic thinking to drive technological innovation and business growth in a dynamic global landscape.		✓	✓	
PEO2. Graduates exemplify integrity and professionalism in their respective fields of specialization, upholding ethical standards, advocating inclusivity, and harnessing technology to drive innovation and meaningful societal impact.	✓	✓	✓	✓
PEO3. Graduates embody a commitment to lifelong learning by continuously enhancing their expertise through relevant certifications, advanced education, and active professional engagement, ensuring adaptability to emerging technologies and industry innovations.	✓	✓	✓	





Program Outcomes (PO)

On completion of the course, the student is expected to be able to do the following:

PROGRAM OUTCOMES	Program Educational Objectives		
	PEO1	PEO2	PEO3
a. Apply computing, science, and mathematics principles to solve discipline-specific problems.	✓		✓
b. Understand and apply industry standards and best practices effectively.	✓	✓	✓
c. Analyze complex problems, identify computing requirements, and devise appropriate solutions.	✓		✓
d. Identify and analyze user needs to guide the selection, creation, evaluation, and management of computer-based systems	✓		✓
e. Design, implement, and evaluate computer-based systems, processes, or programs to meet specific needs under realistic constraints.	✓		✓
f. Deploy IT-based solutions effectively within user environments to enhance usability and performance.	✓		✓
g. Apply current tools, techniques, and practices essential for the IT profession.	✓		✓
h. Function effectively as a member or leader of a development team, recognizing and fulfilling distinct team roles to achieve project goals.	✓	✓	
i. Assist in developing and managing comprehensive IT project plans.	✓	✓	✓
j. Communicate effectively with diverse audiences using logical writing, impactful presentations, and clear technical documentation.	✓	✓	✓
k. Evaluate the local and global implications of IT on individuals, organizations, and society.		✓	✓
l. Understand and adhere to ethical, legal, security, and social responsibilities in IT practice.		✓	✓



m. Recognize the need for self-directed learning and performance improvement to support ongoing professional development.			✓
n. Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.		✓	

Course Outcomes and Relationship to Program Outcomes

Course Outcomes	Program Outcomes													
After completing this course, the student must be able to:	a	b	c	d	e	f	g	h	i	j	k	l	m	n
CO1.Internalize the VMGO of the college especially the discipline towards the achievements of professional competencies, integrity, moral and ethical values.											I	D	D	D
CO2. Appreciate the importance of quantitative methods in decision making.	I		I	I					I					
CO3. Formulate a simplified description of a suitable real-world problem as a mathematical model	E		E	E	E									
CO4. Construct linear programming model and apply solution methods for both maximization and minimization problem.	D		D		E		E		E					
CO5. Explain transportation model algorithms and recognizes transportation model solution methods	E		E		E		E		E					
CO6. Formulate and solve problems as networks and graphs for shortest path, spanning tree,critical path used	D		D		E		D		D					



for planning, scheduling, and controlling project activities.															
CO7. Apply statistical techniques for collection and analysis of data	D	E	D	E	E		D			E					

Note: (I) Introductory Course to an Outcome (E) Enabling Course to an Outcome (D) Demonstrative Course to an Outcome

Course Alignment with the United Nations Sustainable Development Goals

United Nations Sustainable Development Goals	
1. No Poverty	☐ Include topics on economic inequality, poverty alleviation, and inclusive economic growth.
2. Zero Hunger	☐ Discuss food security, sustainable agriculture, and nutrition systems.
3. Good Health and Well-Being	☐ Cover health promotion, mental well-being, and universal healthcare systems.
4. Quality Education	☒ Integrate equitable access to education, lifelong learning, and inclusive teaching practices.
5. Gender Equality	☐ Highlight issues of gender equity, women's empowerment, and reducing gender-based discrimination.
6. Clean Water and Sanitation	☐ Address water resource management, sanitation systems, and access to clean water.
7. Affordable and Clean Energy	☐ Explore renewable energy sources, energy efficiency, and sustainable energy solutions.
8. Decent Work and Economic Growth	☐ Focus on employment opportunities, workplace safety, and sustainable economic policies.
9. Industry, Innovation, and Infrastructure	☒ Discuss sustainable infrastructure, industrialization, and innovation.
10. Reduced Inequalities	☐ Analyze policies and practices that address income and opportunity disparities.
11. Sustainable Cities and Communities	☐ Examine urban planning, disaster resilience, and sustainable living solutions.
12. Responsible Consumption and Production	☐ Address waste reduction, sustainable consumption, and ethical production practices.
13. Climate Action	☐ Discuss mitigation and adaptation strategies for climate change.
14. Life Below Water	☐ Explore marine conservation, sustainable fisheries, and ocean health.
15. Life on Land	☐ Cover biodiversity conservation, reforestation, and land-use practices.
16. Peace, Justice, and Strong Institutions	☐ Promote social justice, conflict resolution, and strong governance.
17. Partnerships for the Goals	☐ Include topics on global collaboration, partnership-building, and cross-sectoral initiatives.





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OTHER REQUIREMENTS AND ASSESSMENTS:

- Active class participation
- Small Group Discussion
- Capstone Project Presentation
- Softcopy and Hardcopy of the Capstone Project
- Capstone Project IMRAD format

GRADING SYSTEM:

Term Examinations	30%
Quizzes	20%
Projects/Outputs/Activities	20%
Participation/Recitation	20%
Class Participation/Attendance	10%
TOTAL	100%

Range	Grade
96.50-100.00	1.00
93.50-96.49	1.25
90.50-93.49	1.50
87.50-90.49	1.75
84.50-87.49	2.00
81.50-84.49	2.25
78.50-81.49	2.50
75.50-78.49	2.75
75.00-75.49	3.00
Conditional Passed	4.00
74 and below (Failed)	5.00
Incomplete	INC
Officially Drop	D
Unofficially Drop	UD
Failure due to Absences	FDA



Final Grade = Midterm Grade + Tentative Final Grade Period
2

LEARNING EPISODES:

Intended Learning Outcomes	Topics	Week	Learning Activities	Suggested Assessment	Plan of Action in Case of Alternative Delivery Mode
Internalize the VMGO of the college especially the discipline towards the achievements of professional competencies, integrity, moral and ethical values.	Class Orientation VMGO of the College Course Outline Grading System Class Policies	1	Memorization /Reflection of VMGO	Recitation	Synchronou s/ Asynchrono us Activity
Demonstrate knowledge in writing the project context, literature and system reviews, and the technical specifications of the study on focus.	Chapter 1: INTRODUCTION • Project Context • Purpose and Descriptio n • Objectives of the Study Chapter 2: REVIEW OF RELATED LITERATURE AND STUDIES • Related	 2-3	Refinement of Chapters 1 to 3 Submission of Revised Chapters 1 to 3	Checking the Revised Chapters 1 to 3	Synchronou s/ Asynchrono us Activity



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	Studies and Systems • Conceptual Framework Chapter 3: TECHNICAL BACKGROUND • Methodology for Analytics Modeling • Methodology for System Development • Requirements Analysis • Testing				
Show competencies in a specific BSIT track as reflected in the creative writing of the result and discussion of the study. Demonstrate knowledge in the writing of the summary of findings, conclusions, and recommendations of the study.	CHAPTER IV: RESULTS and DISCUSSION Discussion of Answers to the Research Objectives Technology Acceptance Model (TAM) ISO 25010 Standards CHAPTER V: SUMMARY and RECOMMENDATION	4-8	Group Dynamics Brain Storming Data Collection and Loading Data Pre-processing Model Development System Development Submission of initial	Checking the initial writeups in Chapters 1 to 3	Synchronous/ Asynchronous Activity



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	Summary of the Research Recommendations		output for chapters 4 and 5		
	Midterm Examination	9	Submission of Technology Acceptance Model (TAM) Questionnaire		Synchronous/ Asynchronous Activity
Demonstrate knowledge in writing the project context, literature and system reviews, and the technical specifications of the study on focus. Show competencies in a specific BSIT track as reflected in the creative writing of the result and discussion of the study. Demonstrate knowledge in writing the summary of findings, conclusions, and recommendations of the study.	System Checking Contents of Chapters 1 to 5	11-13	Refinement of the Developed System System Testing Refinement of Chapters 1 to 5	Mock Presentation	





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Perform high-quality tasks required by the organization, comply with prescribed standards and develop documentation to support it.	Final Presentation	14	Final Presentation	Final Presentation	Synchronou s/ Asynchrono us Activity
Perform high-quality tasks required by the organization, comply with prescribed standards and develop documentation to support it.	Panels’ Recommendati ons	15-16	Complianc e with the panels’ recommen dation	Checking the compliance with the recommenda tion	Synchronou s/ Asynchrono us Activity
Perform high-quality tasks required by the organization, comply with prescribed standards and develop documentation to support it.	IMRAD format of Research Plagiarism Checking	17	Conversion of the final capstone paper to IMRAD format Securing passing plagiarism certificate	Checking the IMRAD equivalent of the Research paper and the compliance with the recommende d acceptable plagiarism level	Synchronou s/ Asynchrono us Activity
Perform high-quality tasks required by the organization, comply with prescribed standards and develop	Capstone Project Hardbound	18	Submission of Capstone Project Hardbound	Checking of complianc e	





documentation to support it.					
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FINAL COURSE OUTPUT:

The students are required to complete a final capstone project that focuses on the design and implementation of data and business analytics applications and other ICT solutions to address the demands in existing business processes.

RUBRIC FOR ASSESSMENT:

Final Defense Rubrics

Oral Presentation 20%	Visual Presentation 10%	Project Output 50%	Response to Questions from Thesis Committee 20%
Presentation delivered in a highly professional manner; confident in material and able to communicate principles clearly; precise diction and syntax; clear command of Standard English.	All supporting visual aids (slides, PowerPoint, etc.) were designed in a highly professional manner; visual aids supported the oral presentation closely, and were clear, concise, and necessary.	The project/system exceeds the requirement on the last proposal defense.	Responds incisively and directly to the questions asked. Responses to questions are specific, defensible, and complex.
Presentation was coherently arranged; scientific principals and results were effectively communicated; occasionally difficult to follow or awkward; some wordiness.	Visual aids were generally well-designed, and communicated the information desired; Some of the visual aids were unnecessary and could have been eliminated.	The project/system meets the requirement on the last proposal defense.	Most responses are direct and relevant to the questions asked. Responses to question are more general, but still accurate; analyses goes beyond the obvious.



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Most scientific principals and results were adequately communicated; much of oral presentation was not adequately prepared or irrelevant; occasional grammatical errors, imprecise diction or awkward syntax; general wordiness.	Visual aids were only adequately designed and often were not able to communicate the information desired; Many of the visual aids were unnecessary and could have been eliminated.	The project/system almost meet the requirement on the last proposal defense.	Responds adequately to the questions asked; occasionally responds with unrelated information. Responses to questions are overly general and disorganized; may have some factual, interpretive, or conceptual errors.
Oral presentation generally confusing; repetitive, wanders; frequent grammatical errors, imprecise diction; wordiness and awkward syntax.	Visual aids were poorly designed and confusing; Many of the visual aids were irrelevant and should not have been used.	The project/system barely meets the requirement on the last proposal defense.	Confuses some significant concepts in the questions asked. Responses to questions are vague or irrelevant
Oral presentation not understandable. Not able to communicate general concepts, results and findings.	Visual aids were sloppy and could not be read or interpreted by the audience.	Did not meet the required output.	Does not understand questions and/or concepts. No discernable response to most questions given.

References:

1. **Johnson, J., & Smith, L. (2022).** *The Capstone Project: A Practical Guide for Students.* Routledge.

2. **Jones, R. A. (2021).** *Completing Your Capstone Project: A Step-by-Step Approach.* SAGE Publications.

3. **Williams, P., & Davis, M. (2023).** *Capstone Success: A Business and Management Guide.* Cengage.

Online Resources:

https://owl.purdue.edu/owl/research_and_citation/apa_style/apa_style_introduction.html

<https://www.citationmachine.net/apa>

<https://sci-hub.se/>

<https://worldwidescience.org/>

Class

Policies:

sarmiento@bulsu.edu.ph | 919-7800 local 1120
Bulacan State University Main Campus, Capitol Compound, McArthur Highway,
Guinhawa, City of Malolos Bulacan





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1. The University rules and regulations will be observed.
2. The students' listed in the master list from the MIS office shall be permitted to attend the class.
3. Enrolled students must, as much as possible attend the face to face classes.
4. Students are expected to take all examinations on the date scheduled and participate actively in the discussion as well as the different activities involved on the subject and the college as well.
5. In case of illness, emergencies, etc., students are required to give an excuse letter signed by the respective guardian. A valid ID must also be presented for validation of the signature.
6. Cheating is equivalent to a lower grade to a failing grade in the subject.
7. Requirements e.g., project, term paper, case study, etc. not submitted on or before the due date will no longer be accepted.
8. Incomplete/Failing grade is given if the student fails to attend the final oral defense.

Prepared by:

Dr. Joann G. Perez
Faculty

Dr. Elenita T. Capariño
Faculty

Checked by:

Jamaica T. Hernandez, MIT
Program Coordinator

Dr. Joann G. Perez
Program Chair

Approved by:

DR. JOFREY R. CAMPOS
Campus Dean



Declaration

I have read and understood the above syllabus in full and in participating in this course I agree to the above rules. I have a clear understanding of the policies and my responsibilities, and I have discussed everything unclear to me with the professor.

I will adhere to the academic integrity and policy and I will treat my fellow students and my professor with due respect.

I understand that this syllabus can be modified or overruled by announcements of the professor in class or on any social media site at any time

_____	_____	_____
Student's Printed name	Signature	Date
_____	_____	_____
Parent's Printed name	Signature	Date

Student's Copy

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_____	_____	_____
Student's Printed name	Signature	Date
_____	_____	_____
Parent's Printed name	Signature	Date

Instructor's Copy